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Name.....

Reg. No.....

FIRST SEMESTER B.Sc. DEGREE EXAMINATION, JANUARY 2012

(CCSS)

Physics—Core Course

PH 1B 01—METHODOLOGY OF SCIENCE AND PHYSICS

Time : Three Hours

Maximum : 30 Weightage

Section A

Answer **all** *questions.*
Each question carries ¼ weightage.

1. Which of the following frames of reference is non-inertial ?

   (a) A space ship cruising uniformly.

   (b) A boat is moving with constant speed.

   (c) An electron revolving uniformly around nucleus.

   (d) An apple at rest on a table.

2. Two photons are moving in opposite direction along same straight line. The relative velocity of one Photon with respect to other Photons is :

   (a) o. (b) 2c.

   (c) c. (d) $\frac{c}{2}$.

3. The degree of freedom of ideal mono atomic gas molecule is :

   (a) one. (b) two.

   (c) 4. (d) 3.

4. The frequencies of X - rays and U.V. radiations are respectively a, b and c, then :

   (a) $a < b < c$. (b) $a < b, b > c$.

   (c) $a > b, b > c$. (d) $a > b, b < c$.

5. The rest mass of a photon is :

   (a) $\frac{hv}{c^2}$. (b) $\frac{hv}{c}$.

   (c) hv. (d) zero.

Turn over

6. A metallic surface ejects photoelectrons when hit by green light, will the electrons be ejecte same surface is hit by red light :

(a) Yes.

(b) Yes if red beam is sufficiently intense.

(c) Yes if red beam fall for enough time.

(d) No.

7. The wavelength of De Broglie wave is independent of :

(a) Charge. (b) Momentum.

(c) Velocity. (d) Mass.

8. The life time an atom in metastable state is :

(a) 10 S. (b) 10^{-8} S.

(c) 10^{-3} S. (d) 10^{3} S.

9. The flux leaving any closed surface per unit volume in a vector field A is called :

(a) grad A. (b) div A.

(c) curl A. (d) flux A.

10. Which of the following operation not defined in vectors?

(a) Division. (b) Multiplication.

(c) Addition. (d) Subtraction.

11. Which among these is not science?

(a) Determining frequency of a musical note.

(b) Cooking food in a pressure cooker.

(c) Distinguishing different smell.

(d) Spray pointing.

12. A positively charged electrons is :

(a) Positron. (b) Proton.

(c) electron. (d) Neutron.

$(12 \times \frac{1}{4} = 3$ weight

Section B

Answer **all** *nine questions.*

13. Briefly explain about 'twin paradox'.

14. What is Compton effect ? Mention its significance.